

Dynamic and static UWB comparison

FlexTDOA, Native DS-TWR and Passive DS-TWR

Final field campaign of August 12, 2026

Platform: 5 ESP32-S3 + DW3000 modules, Raspberry Pi collector
Topology: M1 tag; M2, M3, M4 and M5 anchors
Radio: UWB channel 9; configured 40 MHz SPI
Firmware branch: `localization-raw-calibration`
Evidence: six complete immutable JSONL captures; raw and position-only EKF

Main result. All three protocols produced complete raw dynamic and static datasets. Independent dynamic rates are approximately 25.0 Hz Flex, 25.4 Hz Native and 22.6 Hz Passive. Raw static RMS is 2.3–2.4 cm. The position-only adaptive EKF materially reduces static jitter and dynamic sample-to-sample spikes, while raw UWB remains the accuracy reference.

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1 Executive summary

Evidence boundary. Stream rates, uptime gaps, sample counts, fusion acceptance and reset counts are direct diagnostics. Absolute UWB-RTK disagreement is only indicative: RTK was not an independent survey, clocks are associated in software, antenna lever arms are unknown, and the UWB frame is registered through RTK anchor geometry.

Passive DS overlapping raw windows remain visible in the telemetry rate, while all comparative precision and trajectory metrics use independent frames. Interleaved records marked `imu_fused=true` are excluded from raw UWB metrics. Outliers are retained.

The dynamic independent rates were **25.01 positions/s** for FlexTDOA, **25.35 positions/s** for Native DS-TWR, and **22.63 positions/s** for Passive DS-TWR. Passive DS emitted 70.11 dashboard events/s, but those include overlapping windows.

The static raw local RMS values were **2.41 cm** (FlexTDOA), **2.45 cm** (Native DS-TWR) and **2.26 cm** (Passive DS-TWR). These are precision around each capture median, not absolute accuracy.

The stationary tag RTK centers differed between blocks and the anchor rigid-fit residual is poor in several captures, invalidating an absolute cross-protocol RTK ranking. The new filter uses positions only: this campaign intentionally does not claim acceleration-derived position improvement.

2 Scope and questions

The campaign asks how many independent positions each protocol delivers, where stream gaps occur, how stable the unfiltered stationary position is, and whether the position-only adaptive EKF reduces visible jitter without hiding raw UWB.

The July 26 report used known static geometry and could make a direct accuracy statement. Here the anchors and tag are tied to mobile RTK-derived geometry and software time association, so the evidence boundary is deliberately stricter.

3 System and protocol background

3.1 FlexTDOA

The tag is UWB receive-only. Anchor transmissions form directed time-difference observations and the tag solves from a recent coherent observation set. Radio airtime is largely independent of the number of passive tags.

3.2 Native DS-TWR

The active tag exchanges poll, response and final messages with every anchor, obtains four absolute ranges and solves a coherent frame. Airtime grows with active tag count, but frame semantics are direct.

3.3 Passive DS-TWR

The tag listens to anchor-to-anchor DS-TWR exchanges. Rolling windows may produce several publications from shared measurements, so dashboard events and independent frames are intentionally reported separately.

4 Test configuration and captured evidence

The three operator-marked dynamic walks lasted 85–114 s and the stationary blocks approximately 30 s. The selector uses the final continuous-capture segments and the final short static capture for each protocol.

Protocol	Mode	Capture ID	duration s	events	independent	indep Hz	coverage pct	complete
FlexTDOA	dynamic	flextdoa_final_dynamic_20260812	84.6	2117	2117	25.01	99.7	yes
Native DS-TWR	dynamic	native_ds_final_dynamic_20260812	113.5	2876	2876	25.35	98.7	yes
Passive DS-TWR	dynamic	passive_ds_final_dynamic_20260812	106.5	7463	2409	22.63	100.0	yes
FlexTDOA	static	flextdoa_static_final.01	30.3	798	798	26.30	98.8	yes
Native DS-TWR	static	native_ds_static_final.02_rtk_fixed	30.1	780	780	25.89	99.6	yes
Passive DS-TWR	static	passive_ds_static_final.01	31.2	1906	607	19.45	96.2	yes

Every selected block has a terminal `capture_end`. All raw data are archived losslessly and identified by SHA-256.

5 Methodology

5.1 Population and continuity

No error-based trimming or temporal smoothing is applied. Rates use full capture wall time. Gaps are positive differences between unique ESP32 uptime values; P50/P95/P99, maximum and counts above 100 ms are retained together.

5.2 Static precision

Radial displacement is computed around each block's component-wise position median. CEP50, RMS, P95, P99, maximum, 2DRMS and first-to-last-decile drift use all independent samples. They describe repeatability, not offset from truth.

5.3 RTK audit

Only quality-4 fixes enter the audit. WGS84 is converted through ECEF to a common ENU frame. A proper 2-D rotation and translation is fitted from at least three anchor centers; scale is fixed. Tag matches must be within 100 ms. Anchor-fit residual and association timing accompany every tag error.

6 Dynamic results

6.1 Raw trajectories

The walks are separate physical trials, not synchronized repeats. The panels therefore show continuity and gross geometry rather than point-by-point route agreement.

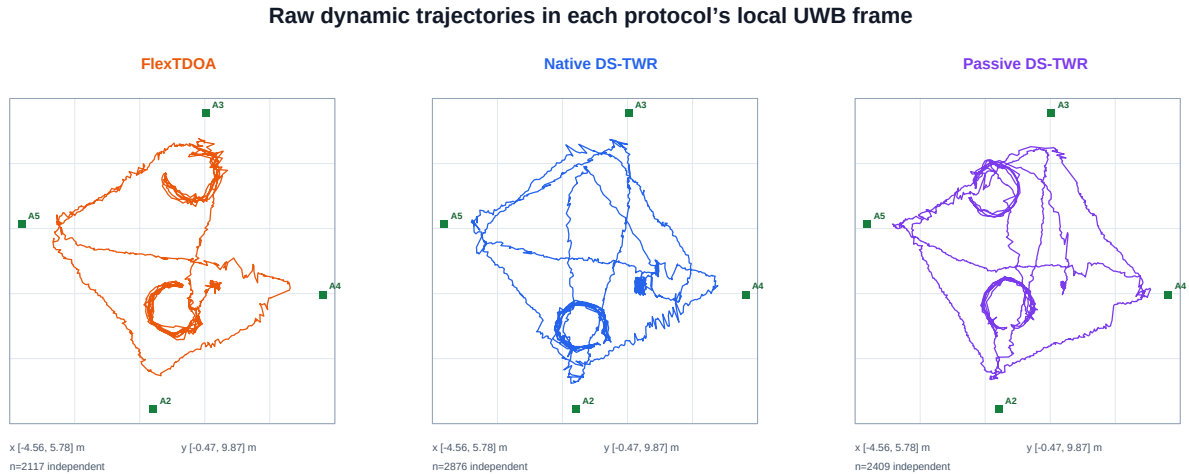


Figure 1: Raw independent dynamic trajectories in each local UWB frame. Green markers are anchors. Different panels represent different walks.

6.2 Update rate and gaps

Protocol	duration s	events	independent	event Hz	indep Hz	gap P95 ms	gap max ms	step P95 m/s
FlexTDOA	84.6	2117	2117	25.01	25.01	70	830	6.04
Native DS-TWR	113.5	2876	2876	25.35	25.35	50	130	7.15
Passive DS-TWR	106.5	7463	2409	70.11	22.63	30	250	4.66

Native DS-TWR delivered the highest independent rate (25.35 Hz) and the best worst-case continuity (130 ms). FlexTDOA was close in rate (25.01 Hz), but its maximum gap was 830 ms. Passive DS delivered 22.63 independent Hz despite overlapping dashboard publications, with a 250 ms maximum gap. Step speed is a sample-to-sample trajectory diagnostic, not measured walking speed.

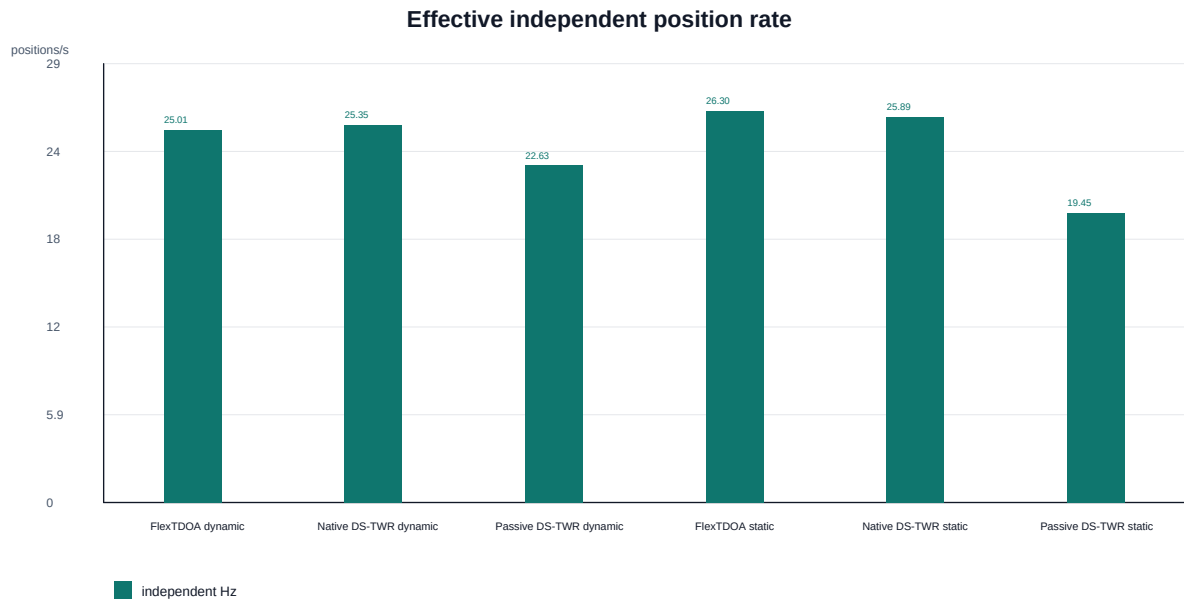


Figure 2: Independent position rate in the selected dynamic and static blocks. Passive DS all-event publications are not substituted for independent frames.

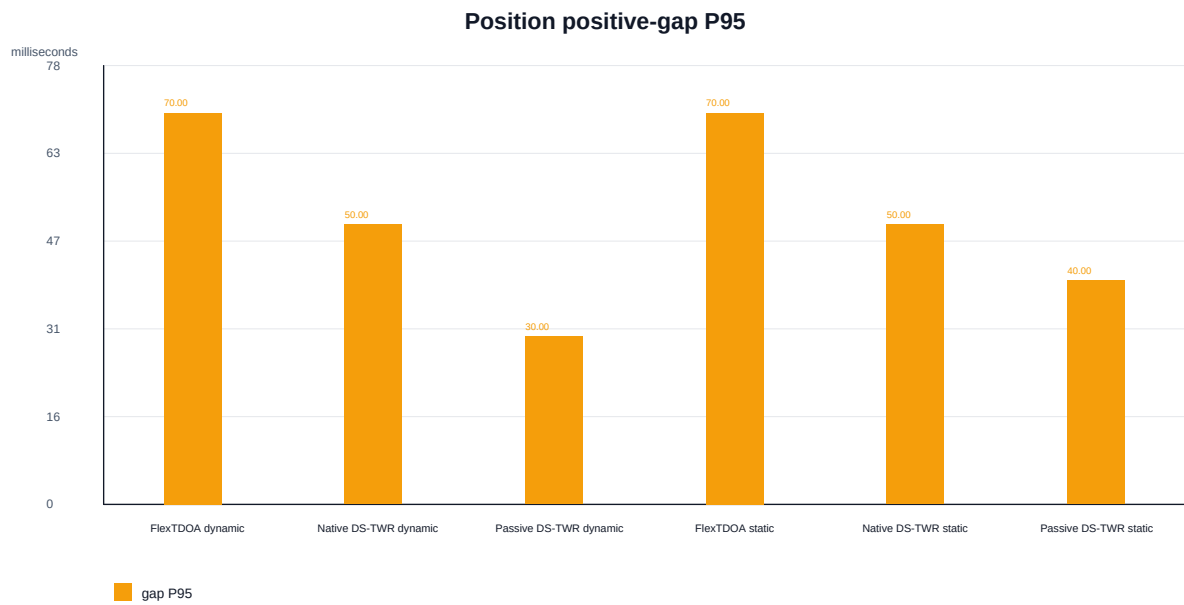


Figure 3: P95 positive gap between unique position uptimes. Maximum gaps and counts remain in the tables.

6.3 Complete continuity audit

Protocol	Mode	event Hz	indep Hz	P50 ms	P95	P99	max	± 100 ms
FlexTDOA	dynamic	25.01	25.01	30	70	130	830	28
Native DS-TWR	dynamic	25.35	25.35	40	50	70	130	2
Passive DS-TWR	dynamic	70.11	22.63	10	30	80	250	28
FlexTDOA	static	26.30	26.30	30	70	100	120	7
Native DS-TWR	static	25.89	25.89	40	50	60	80	0
Passive DS-TWR	static	61.06	19.45	10	40	72	130	6

7 Position-only adaptive EKF

The dashboard EKF uses only the raw UWB position stream. IMU acceleration is disabled, raw measurements remain archived separately, and each EKF result in this report comes from the same position event as its raw counterpart. Sample-to-sample step-speed P95 is a jitter/continuity diagnostic; it is not physical walking speed. Static RMS is repeatability about each stream's own median. RTK values are audit-only whenever anchor registration is poor.

Protocol	Mode	coverage pct	raw step P95	EKF step P95	raw static RMS cm	EKF static RMS cm	raw RTK m	EKF RTK m
FlexTDOA	dynamic	100.0	6.04	4.31	n/a	n/a	1.610	1.609
Native DS-TWR	dynamic	100.0	7.15	3.44	n/a	n/a	1.694	1.710
Passive DS-TWR	dynamic	100.0	4.66	4.50	n/a	n/a	1.397	1.395
FlexTDOA	static	100.0	1.97	0.36	2.41	1.07	0.599	0.599
Native DS-TWR	static	100.0	1.61	0.23	2.45	0.89	n/a	n/a
Passive DS-TWR	static	100.0	1.62	0.00	2.26	0.00	n/a	n/a

Dynamic raw versus position-only adaptive EKF

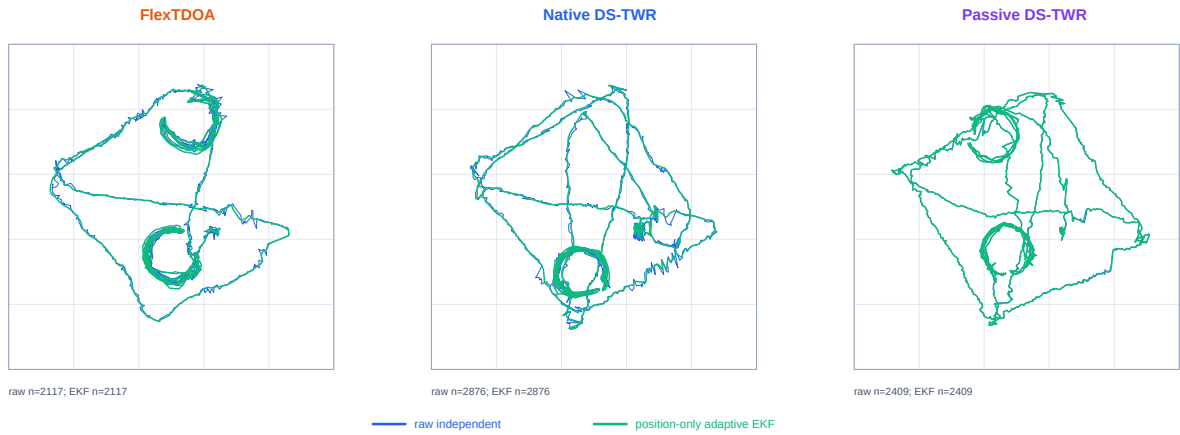


Figure 4: Raw independent dynamic trajectory (blue) and the position-only adaptive EKF (green). The EKF reduces high-frequency corners without replacing archived raw data.

8 Static results

8.1 Raw local precision

Protocol	N	CEP50 cm	RMS cm	P95 cm	P99 cm	max cm	2DRMS cm	drift cm
FlexTDOA	798	1.91	2.41	4.30	5.73	7.38	4.81	0.82
Native DS-TWR	780	1.72	2.45	3.95	8.81	12.30	4.84	0.67
Passive DS-TWR	607	1.88	2.26	3.94	4.93	5.52	4.53	0.38

All raw static RMS values lie in a narrow 2.26–2.45 cm range. Passive DS is best on RMS, while Native and Passive share the best P95 near 3.94 cm. All outliers are retained. One stationary point cannot characterize GDOP or spatial bias across the anchor polygon.

The RTK Fixed clouds are individually useful, but their common-ENU centers do not agree across blocks. High short-term RTK precision therefore does not imply cross-session correctness.

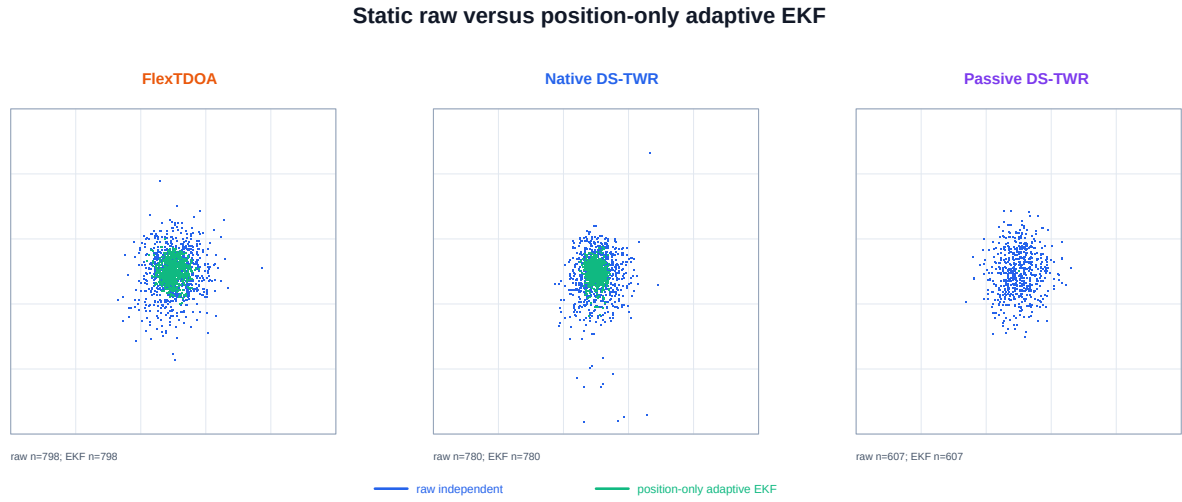


Figure 5: Static raw and EKF clouds, each centered on its own median and plotted at one common scale. Passive DS may collapse to a held point while stationary by design.

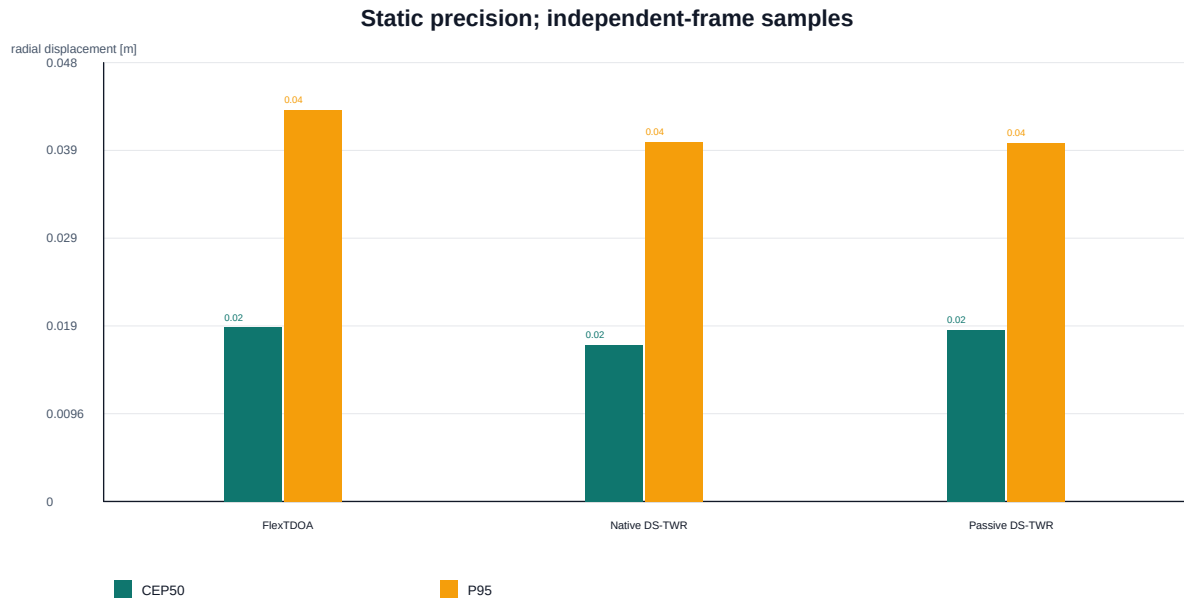


Figure 6: Static precision around each block's own median. All independent raw samples are included.

Static raw position clouds about each capture median

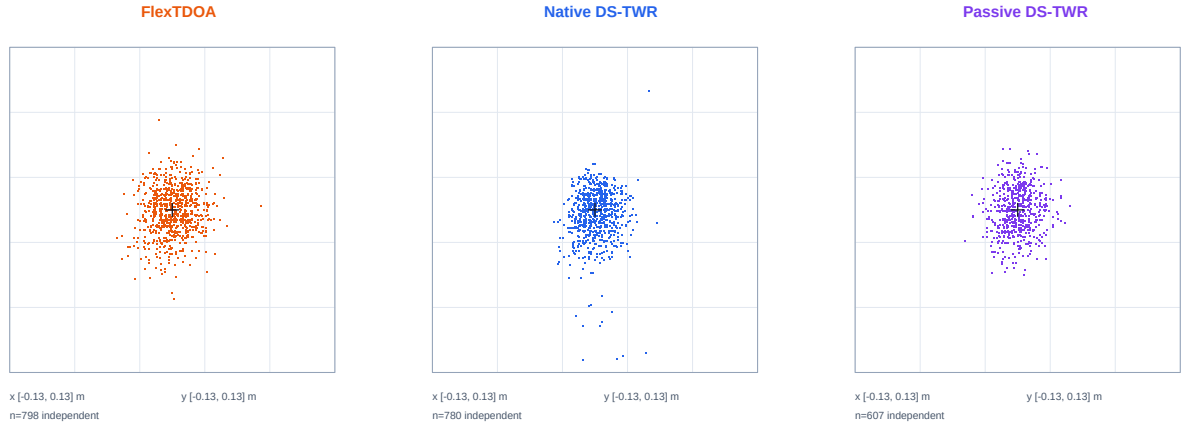


Figure 7: Static raw position clouds centered on their own medians and drawn at one common scale. The cross is the median.

GPS RTK Fixed tag clouds and cross-capture centers

Top: each block centered on its own median. Bottom: the same fixes in one common ENU frame.



Figure 8: GPS RTK Fixed tag solutions. Top: each protocol block is centered on its own median with one shared centimetre scale, showing within-block precision. Bottom: the same fixes and medians in the common ENU frame with equal axis scale, exposing cross-capture reference inconsistency.

8.2 RTK cross-capture consistency gate

The stationary RTK center's maximum pairwise separation is **2.127 m**, versus a 0.250 m gate. **static absolute UWB-RTK errors are audit-only and invalid for cross-protocol ranking.** The failed gate forbids an absolute static ranking even when an individual cloud is tight.

9 Accelerometer scope

Acceleration-assisted positioning was disabled for this final campaign because the preceding experiment did not improve raw UWB accuracy. The adaptive EKF assessed above is position-only. Orientation/IMU telemetry is outside the protocol ranking, and raw UWB remains the permanent audit stream.

10 UWB-RTK association audit

The following values are retained for transparency, not promoted to surveyed accuracy. Debaised RMSE removes median tag residual and is a shape diagnostic.

Protocol	Mode	pairs	time P95 ms	RMSE m	P95 m	debaised m	anchor fit m	alignment
FlexTDOA	dynamic	656	34.9	1.610	1.828	0.376	0.859	poor_alignment_do_not_claim_accuracy
Native DS-TWR	dynamic	633	18.5	1.694	2.391	1.202	0.675	poor_alignment_do_not_claim_accuracy
Passive DS-TWR	dynamic	726	45.8	1.397	1.555	0.486	0.013	usable_with_limitations
FlexTDOA	static	137	33.2	0.599	0.629	0.025	0.930	poor_alignment_do_not_claim_accuracy
Native DS-TWR	static	0	n/a	n/a	n/a	n/a	n/a	unavailable
Passive DS-TWR	static	0	n/a	n/a	n/a	n/a	n/a	unavailable

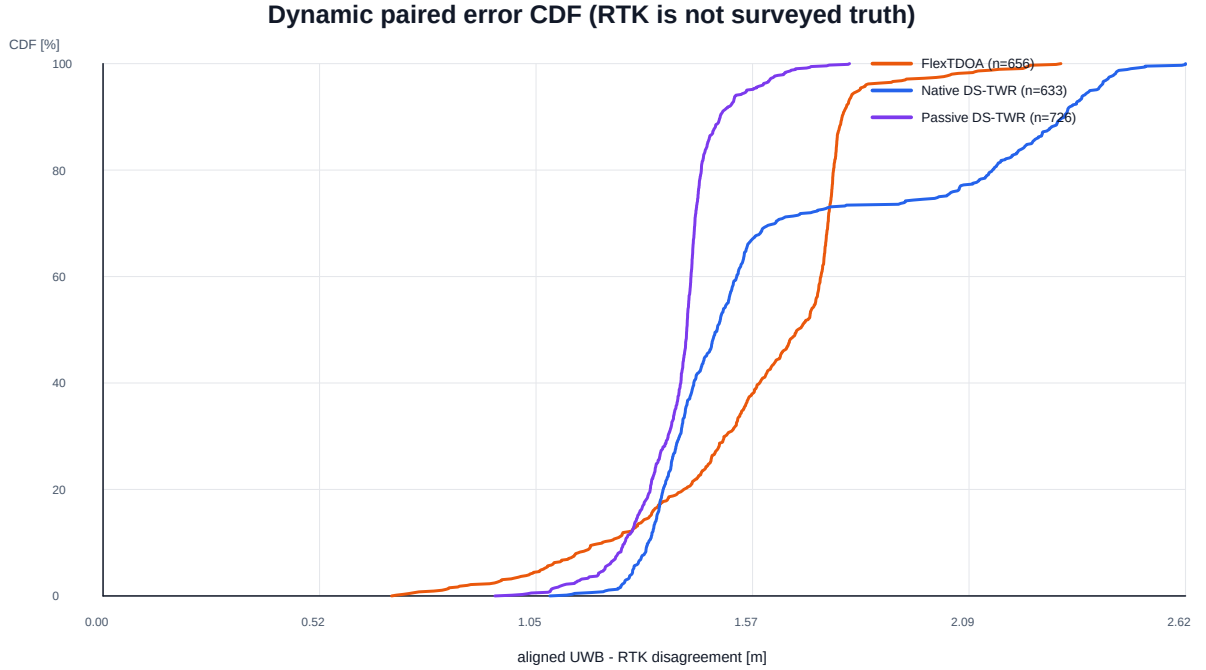


Figure 9: Dynamic UWB-RTK disagreement after anchor-based rigid registration. This is not surveyed position error.

Dynamic paths and anchor fits differ by capture. Passive DS dynamic geometry is partly GNSS-derived, making its registration non-independent. The RTK audit is still useful for gross frame errors and timing failures.

11 Protocol interpretation and selection

11.1 FlexTDOA

Near-25 Hz independent dynamic rate with a receive-only scalable tag. Static raw RMS was 2.41 cm. The 830 ms maximum gap is the principal caveat.

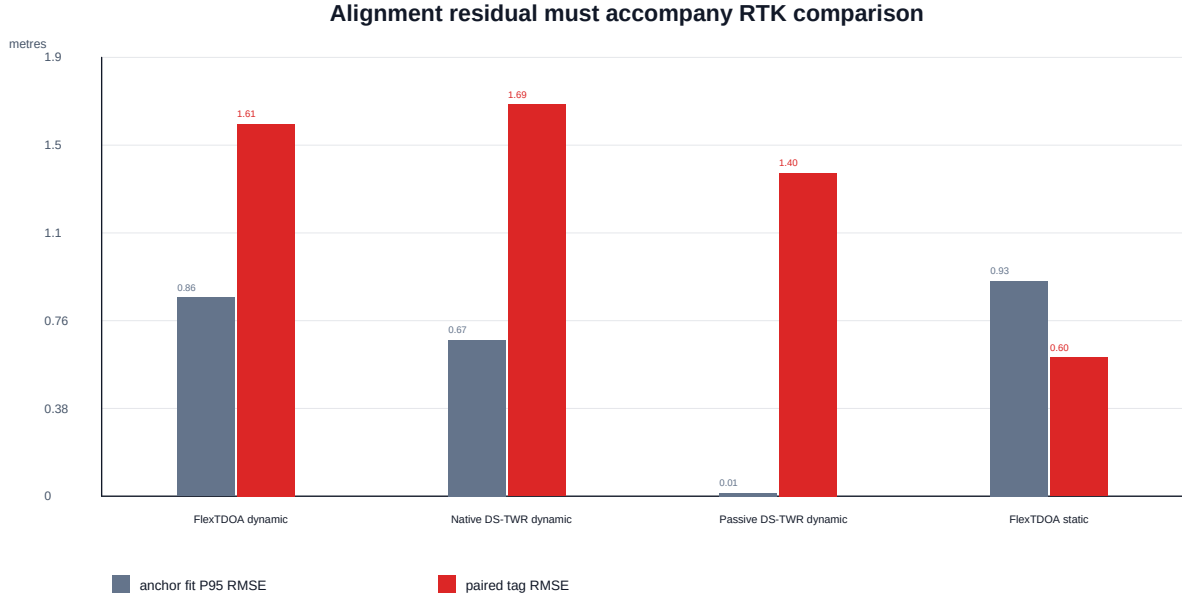


Figure 10: Anchor-registration residual beside paired tag disagreement. Registration uncertainty limits tag interpretation.

11.2 Native DS-TWR

Clearest coherent-frame semantics, highest measured independent rate and best maximum gap. Static raw RMS was 2.45 cm. It remains the preferred deterministic baseline for a small number of active tags.

11.3 Passive DS-TWR

Successfully follows motion without tag transmissions. Independent-frame accounting yields 22.63 Hz despite overlapping publications; static raw RMS was the best at 2.26 cm and the maximum dynamic gap was 250 ms.

Criterion	FlexTDOA	Native DS-TWR	Passive DS-TWR
Tag radio role	receive-only	active exchanges	receive-only
Dynamic independent rate	25.01 Hz	best: 25.35 Hz	22.63 Hz
Dynamic worst gap	830 ms	best: 130 ms	250 ms
Static raw precision	2.41 cm RMS	2.45 cm RMS	best: 2.26 cm RMS
Scaling with tag count	strongest	airtime grows	strong
Recommendation	primary scalable mode	coherent baseline	passive scalable alternative

12 Limitations

- RTK is an in-system comparison reference, not independently surveyed truth.
- Measurement time subtracts receiver-reported fix age from collector time; clocks are not hardware synchronized.
- GNSS and UWB antenna phase centers have unknown lever arms.
- Dynamic routes are not repeated surveyed paths; static/dynamic blocks are single trials.
- One static point does not cover edge geometry, NLOS, orientation or spatial bias.
- Collector rate is end-to-end telemetry, not direct DW3000 PHY throughput.

13 Recommended next tests

1. Repeat every protocol on one marked route with identical orientation and synchronized markers.
2. Interleave static blocks at the center, edges and outside the polygon.

3. Keep one continuous RTK session and log correction age/reset state across protocol changes.
4. Add hardware-correlated timestamps for latency claims.
5. Keep acceleration disabled until body-to-map yaw is independently validated.
6. Capture focused FlexTDOA diagnostics around gaps above 250 ms.

14 Preliminary findings

- FlexTDOA remains a strong receive-only scalable mode, with one long-gap caveat.
- Native DS-TWR is the best coherent-frame rate and continuity baseline.
- Passive DS-TWR is a viable receive-only alternative when independent frames are counted correctly.
- The RTK consistency failure prevents a static absolute-accuracy ranking; local precision alone is ranked.
- The position-only EKF improves display stability; acceleration-assisted accuracy is not claimed.

15 Reproduction and audit files

The directory contains `analysis_summary.json`, capture/replay/alignment CSV files, SVG/PDF plots and this TeX source. The machine-readable summary records whether optional lossless RAW archive generation was enabled; when it is skipped, immutable source JSONL paths remain recorded for reproduction.

```
python tools/uwb_dynamic_static_report.py --input-dir INPUT --output-dir OUTPUT
```

No raw capture is modified during report generation.